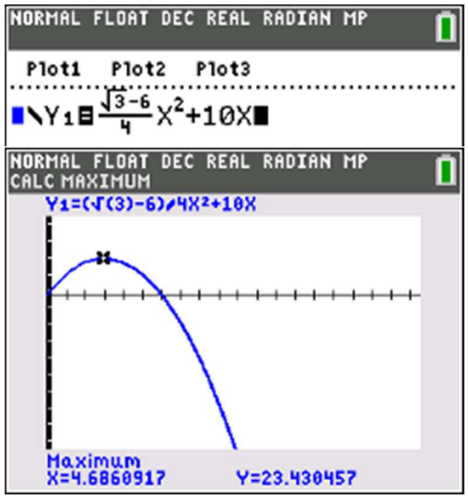
Raffles Institution
H2 Mathematics (9758)
Solution for 2024 A-Level Paper 2

Section A: Pure Mathematics

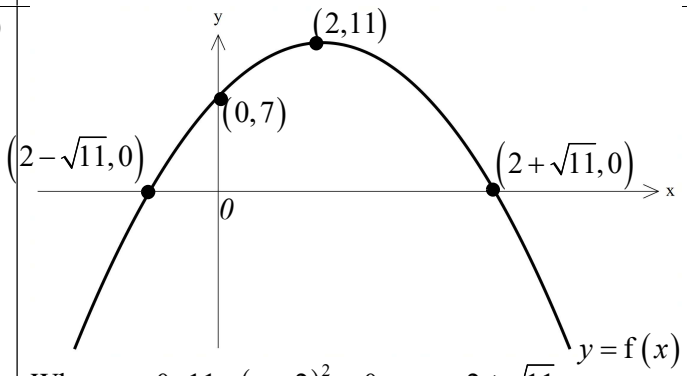
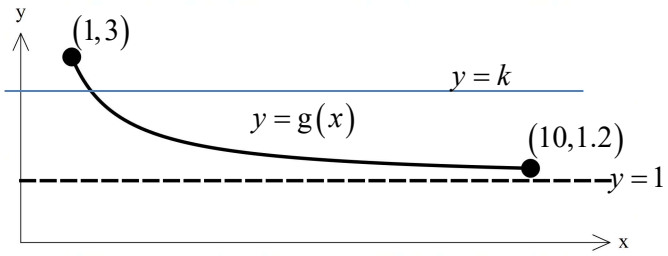
Question 1

	Suggested Solution	Remarks for Student
(a) **[2]**	$z = -2 + \mathrm{i}$ $z = az^* + b$, where $a, b \in \mathbb{Z}$ $\Rightarrow -2 + \mathrm{i} = a(-2 - \mathrm{i}) + b$ Comparing real part, $-2 = -2a + b \quad \cdots(1)$ Comparing imaginary part, $1 = -a \Rightarrow a = -1 \quad \cdots(2)$ Substitute into $(1): b = -4$	Calculator can be used to verify answers obtained.
(b) **[4]**	$w = 1 - 3\mathrm{i}$ $wz - \dfrac{w}{z}$ $= (1 - 3\mathrm{i})(-2 + \mathrm{i}) - \dfrac{1 - 3\mathrm{i}}{-2 + \mathrm{i}} \times \dfrac{-2 - \mathrm{i}}{-2 - \mathrm{i}}$ $= \left[-2 + 3 + \mathrm{i}(1 + 6)\right] - \dfrac{-2 - 3 + \mathrm{i}(-1 + 6)}{4 + 1}$ $= 1 + 7\mathrm{i} - \dfrac{1}{5}(-5 + 5\mathrm{i})$ $= 2 + 6\mathrm{i}$	Detailed working needs to be shown since use of calculator is prohibited in this question (calculator can be used to verify answers).

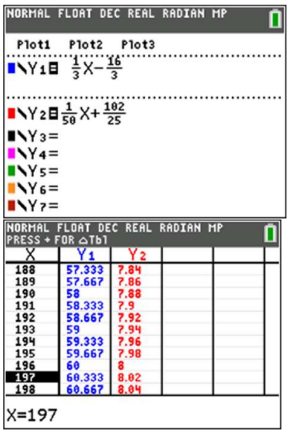
Question 2

	Suggested Solution	Remarks for Student
[6]	Considering perimeter of flower bed: $2b+3a=20 \Rightarrow b=10-\frac{3}{2}a \quad \cdots(*)$ Area, $A=ab+\frac{1}{2}a^2\sin 60^\circ$ Substitute (*): $A=a\left(10-\frac{3}{2}a\right)+\frac{1}{2}a^2\left(\frac{\sqrt{3}}{2}\right)$ $=\frac{\sqrt{3}-6}{4}a^2+10a$ $\frac{\mathrm{d}A}{\mathrm{d}a}=\frac{\sqrt{3}-6}{2}a+10$ When $\frac{\mathrm{d}A}{\mathrm{d}a}=0, \quad \frac{\sqrt{3}-6}{2}a+10=0$ $\Rightarrow a=\frac{20}{6-\sqrt{3}}$	Since A contains a^2 and a, it is more efficient to express A in terms of a only than in terms of b.
	$\frac{\mathrm{d}^2A}{\mathrm{d}a^2}=\frac{\sqrt{3}-6}{2}\approx -2.13<0$ By 2nd derivative test, maximum area occurs when $a=\frac{20}{6-\sqrt{3}}$. $\therefore$ Maximum area is $\frac{\sqrt{3}-6}{4}\left[\frac{20}{6-\sqrt{3}}\right]^2+10\left[\frac{20}{6-\sqrt{3}}\right]$ $=23.43 \text{ m}^2 (4\text{sf})$	You can verify the value obtained for maximum area by using a GC as follows:  Interestingly, with the use of the phrase "shorter sides", it is implied that $a<b$. However, when $a=\frac{20}{6-\sqrt{3}}\approx 4.6861$, $b=10-\frac{3}{2}\left(\frac{20}{6-\sqrt{3}}\right)\approx 2.9709$ which suggests that $a>b$.

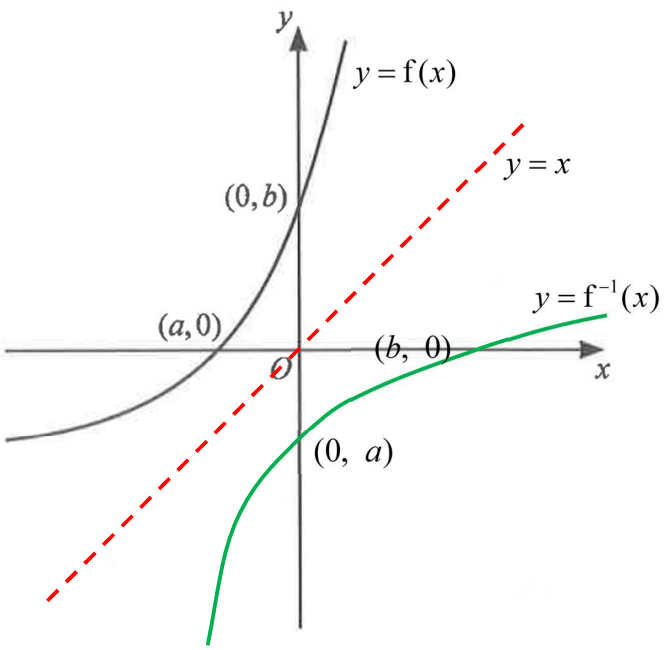
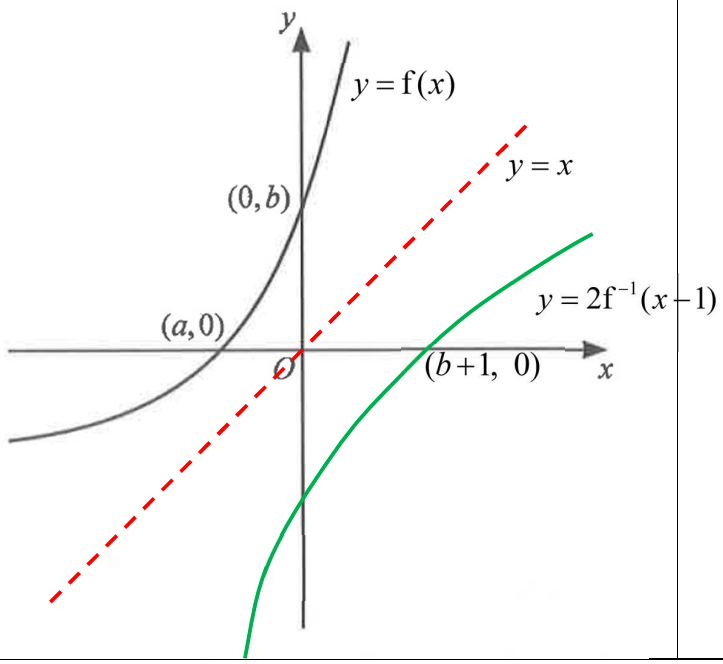
Question 3

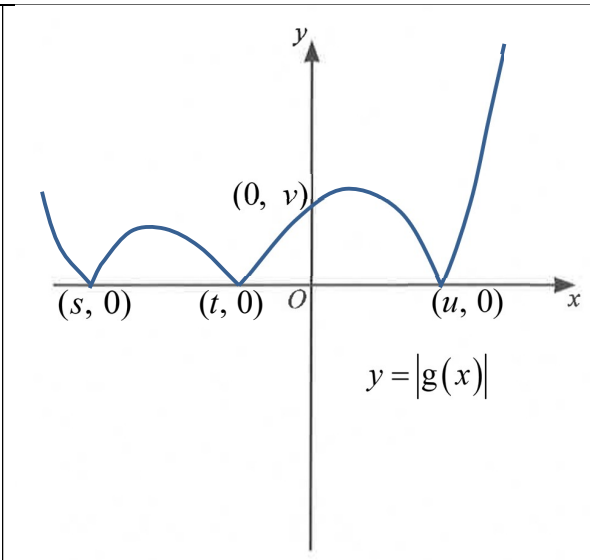
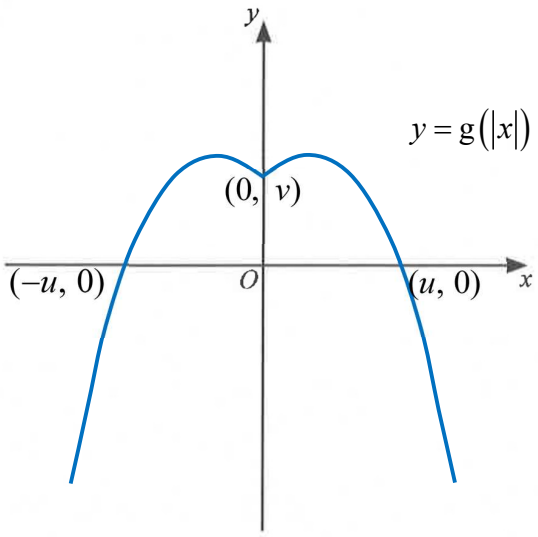
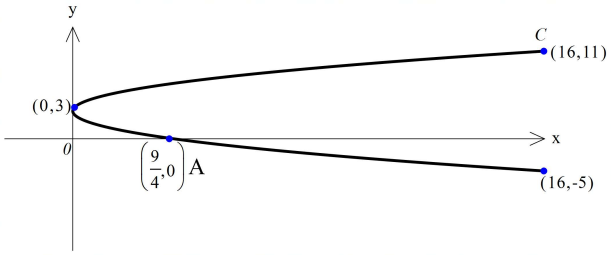
	Suggested Solution	Remarks for Student
(a) [2]	$\mathrm{f}(x) = -x^2 + 4x + 7, x \in \mathbb{R}$ $= 11 - (x-2)^2 \le 11$ Range of f, $\mathrm{R_f} = (-\infty, 11]$	Since this is a quadratic function, we can complete the square to find the range of f.
(b) [2]	 When $y = 0$, $11 - (x-2)^2 = 0 \Rightarrow x = 2 \pm \sqrt{11}$	
(c) [1]	$\mathrm{g}(x) = \frac{2}{x} + 1, \quad 1 \le x \le 10$  Any horizontal line $y = k$, $1.2 \le k \le 3$, cuts the graph of $y = \mathrm{g}(x)$ at one and only one point. Hence, g is one-one and g^{-1} exists.	It is essential to state the range of values of k, where $k \in \mathrm{R}_g$
(d) [3]	Consider $\mathrm{g}(x) = \frac{2}{x} + 1 = 1.5$ $\Rightarrow x = \frac{2}{0.5} = 4 \Rightarrow \mathrm{g}^{-1}(1.5) = 4$ $\mathrm{fg}^{-1}(1.5) = \mathrm{f}(4) = 7$	

Question 4

No.	Suggested Solution	Remarks for Student
(a) **[2]**	For $n \geq 2$, $t_n = \left[2n^3 - 8n^2 - 4n\right] - \left[2(n-1)^3 - 8(n-1)^2 - 4(n-1)\right]$ $= \left[2n^3 - 8n^2 - 4n\right] - \left[2(n^3 - 3n^2 + 3n - 1) - 8(n^2 - 2n + 1) - 4(n-1)\right]$ $= 6n^2 - 22n + 6$ $= 2(3n^2 - 11n + 3)$ For $n = 1$, $t_n = 2 - 8 - 4 = -10 = 2(3 - 11 + 3)$ $\therefore t_n = 2(3n^2 - 11n + 3)$ for $n \geq 1$	
(b) **[2]**	For $u_n = t_n$, $50n - 204 = 6n^2 - 22n + 6$ $\Rightarrow 6n^2 - 72n + 210 = 0$ $\Rightarrow n^2 - 12n + 35 = 0$ $\Rightarrow n = 5$ or 7	
(c) **[2]**	Consider $u_n = 50n - 204$ and $v_m = 3m + 16$ where $u_n = v_m = l$. l must be an integer since m and n are integers. $50n - 204 = l$ and $3m + 16 = l$ $\Rightarrow 50n - l = 204 \cdots (1)$ and $3m - l = -16 \cdots (2)$ Solving, $m = \frac{1}{3}l - \frac{16}{3}$ and $n = \frac{1}{50}l + \frac{102}{25}$. By using GC, the smallest integer l so that both m and n are positive integers is 196 (corresponding to u_8 and v_{60})	
(di) **[2]**	$w_n = 3n^2 - 5n + 7$ When n is odd, $3n^2$ is odd and $5n$ is also odd, so $3n^2 - 5n$ is even, Hence $3n^2 - 5n + 7$ is odd. When n is even, $3n^2$ is even and $5n$ is also even, so $3n^2 - 5n$ is even, Hence $3n^2 - 5n + 7$ is odd. Therefore, all the terms in W are all odd numbers.	
(dii) **[1]**	$u_n = 50n - 24 = 2(25n - 12)$ $\Rightarrow$Terms in U are all even numbers. Since all the terms in W are all odd numbers, the series U and W do not have any terms in common.	

Question 5

	Suggested Solution	Remarks for Student
(ai) **[1]**		Graph of $y=f^{-1}(x)$ is a reflection of graph of $y=f(x)$ about the line $y=x$.
(aii) **[2]**		

(bi) **[2]**		
(bii) **[2]**		There is symmetry about the y axis
(ci) **(cii)** **[1]** **[2]**		Note that $-4 \le t \le 4$, do adjust the window settings accordingly. The two "end" points corresponding to t=±4 needs to be labelled on the sketch.
(ciii) **[1]**	$y = 3$ is a line of symmetry.	

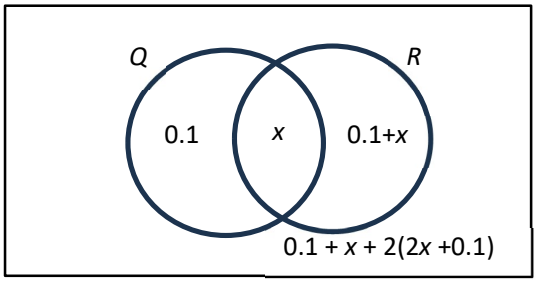
Section B: Probability and Statistics

Question 6

	Suggested Solution	Remarks for Student
(a) **[1]**	Total membership $= 27 + 45 + 18 = 90$ Since $65 < 90$, not all members of the club gave replies for the questionnaire. Hence the 65 members form a sample.	
(b) **[1]**	An advantage of choosing a random sample of members for her committee is that the choice of members would not be biased, or the choice would be fair or the choices would be representative of the whole membership of the club.	Students should avoid stating the following, as they do not address the question: 1) Members chosen would not be biased (note the difference in phrasing) 2) Each member has an equal chance of being selected 3) Members are selected independently
(c) **[4]**	Probability that the committee contains more Youth members than Senior members $= \mathrm{P}(AAAYYS)$ $= \dfrac{\binom{45}{3}\binom{27}{2}\binom{18}{1}}{\binom{45}{3}\binom{27}{2}\binom{18}{1} + \binom{45}{3}\binom{27}{1}\binom{18}{2} + \binom{45}{4}\binom{27}{1}\binom{18}{1}}$ $= \dfrac{89652420}{220682880}$ $= \dfrac{13}{32}$ or 0.40625	To meet the 3 criteria provided by the bulleted points in the question: we can have the following 3 cases: 1) 3 Adult members, 2 Youth members and 1 Senior member, or 2) 3 Adult members, 1 Youth member and 2 Senior members, or 3) 4 Adult members, 1 Youth member and 1 Senior member

Question 7

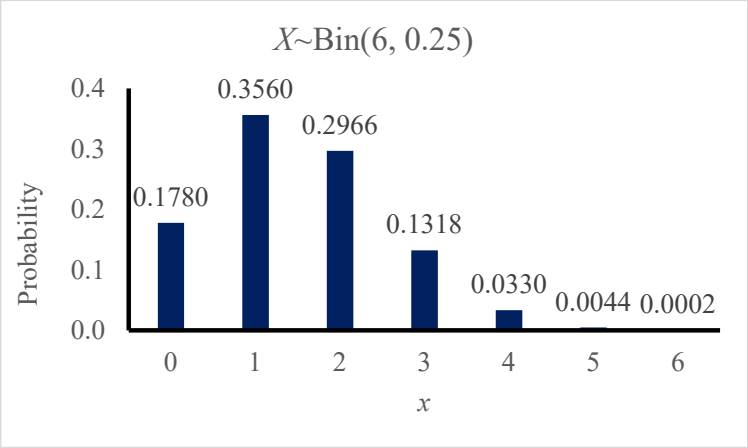
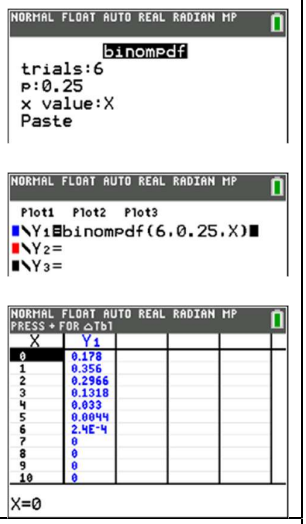
No.	Suggested Solution	Remarks for Student
(ai) [1]	Probability that A finds the prize $=\left(\frac{1}{n}\right)+\left(\frac{n-1}{n}\right)\left(\frac{1}{n-1}\right)$ $=\frac{2}{n}$	
(aii) [1]	Probability that B finds the prize $=\left(\frac{n-1}{n}\right)\left(\frac{n-2}{n-1}\right)\left[\left(\frac{n-3}{n-2}\right)\left(\frac{1}{n-3}\right)+\frac{1}{n-2}\right]$ $=\frac{2}{n}$ Alternatively, Probability that B finds the prize = P(A does not find prize) $\times$ P(B finds prize given that A does not find prize) $=\left(1-\frac{2}{n}\right)\left(\frac{2}{n-2}\right)$ (by applying (ai))	
(aiii) [3]	Probability that C finds the prize $=1-\frac{2}{n}-\frac{2}{n}=1-\frac{4}{n}$ For probability that C finds the prize is more than 8 times the probability that either A or B find the prize, so $1-\frac{4}{n}>8\left(\frac{2}{n}+\frac{2}{n}\right)$ $\Rightarrow 1-\frac{4}{n}>\frac{32}{n}$ $\Rightarrow 1>\frac{36}{n}$ $\Rightarrow n>36$ Therefore, $n\in\{n\in\mathbb{Z}^+ \mid n>36\}$ or $n\in\{n\in\mathbb{Z}^+ \mid n\geq 37\}$	

(b) [3]	 $x+0.1+x+0.1+0.1+x+2(2x+0.1)=1$ $\Rightarrow 7x=1-0.5$ $\Rightarrow x=\frac{1}{14}$ **OR** $\mathrm{P}(Q\cup R)=\mathrm{P}(Q)+\mathrm{P}(Q'\cap R)=0.1+x+0.1+x=0.2+2x$ $\mathrm{P}((Q\cup R)')=1-\mathrm{P}(Q\cup R)=\mathrm{P}(Q)+2\mathrm{P}(R)$ $\Rightarrow 1-(0.2+2x)=(0.1+x)+2(0.1+2x)$ $\Rightarrow 0.8-2x=0.3+5x$ $\Rightarrow 0.5=7x$ $\Rightarrow x=\frac{1}{14}$	Draw a Venn diagram to help in visualisation

Question 8

No.	Suggested Solution	Remarks for Student
(a) **[1]**	Lee should carry out a 1-tail test since he suspects that the mean number of birds visiting the new bird feeder while he eats his breakfast has reduced or decreased.	Answer should be given in context.
(b) **[2]**	Let μ be the **population mean number of birds** visiting the bird feeder during a 20 minute interval each morning. Null hypothesis, $H_0 : \mu = 17.3$ Alternative hypothesis, $H_1 : \mu < 17.3$	Note that question requires us to define "any parameters you use". So ensure that you have defined μ.
(c) **[3]**	An unbiased estimate of the population mean is $\bar{x}$ $= \frac{\sum x}{n} = \frac{512}{32} = 16$ An unbiased estimate of the population variance is s^2 $= \frac{1}{n-1}\left(\sum x^2 - \frac{\left(\sum x\right)^2}{n}\right) = \frac{1}{31}\left(8702 - \frac{(512)^2}{32}\right) = \frac{510}{31}$ Under H_0, since $n = 32$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(17.3, \frac{\frac{510}{31}}{32}\right)$ approximately . Since Lee concluded that the null hypothesis should be rejected, $p\text{-value} = P\left(\bar{X} \le 16\right) \le \alpha\%$ Using GC, p-value = $0.0349106089 \le \frac{\alpha}{100}$ $\Rightarrow \alpha \ge 3.49106$ Therefore, the minimum possible value of the integer α is 4.	
(d) **[1]**	Since p-value = $0.347735667 \le \alpha\%$, we **reject** H_0 and conclude that there is **sufficient** evidence at $\alpha\%$ level of significance that the <u>mean</u> number of birds visiting the new bird feeder while Lee eats his breakfast has reduced, in other words, Lee's suspicion is correct.	
(e) **[1]**	Using the number of birds for each of 10 mornings would not be sufficient for Lee to carry out his hypothesis test as the distribution of the number of birds visiting the feeder is unknown and the sample size of 10 mornings is not sufficiently large for Central Limit Theorem to be applied.	

Question 9

	Suggested Solution	Remarks for Student
(a) **[2]**		The chart should show 7 separate bars or lines, labelled 0 to 6 to represent the probabilities $\mathrm{P}(X=0)$, $\mathrm{P}(X=1),\cdots,$ $\mathrm{P}(X=7)$. The heights of the bars should be of correct relative heights. Use GC to get the probabilities: 
(b) **[2]**	Estimated probability that a randomly chosen pie is unsatisfactory is $= \dfrac{(0)(57)+(1)(9)+(2)(6)+(3)(4)+(4)(3)+(5)(1)}{80\times 50}$ $= \dfrac{1}{80}$	
(ci) **[2]**	The two assumptions needed for Y to be well modelled by a binomial distribution are : **1.** The **event** that a randomly chosen burger has insufficient cheese is **independent of** the **event** that another randomly chosen burger has insufficient cheese. **2.** The **probability** of a burger having insufficient cheese **remains constant** for all burgers.	It is incorrect to say that the *probability* of a burger having insufficient cheese is independent of the *probability* of another burger.....

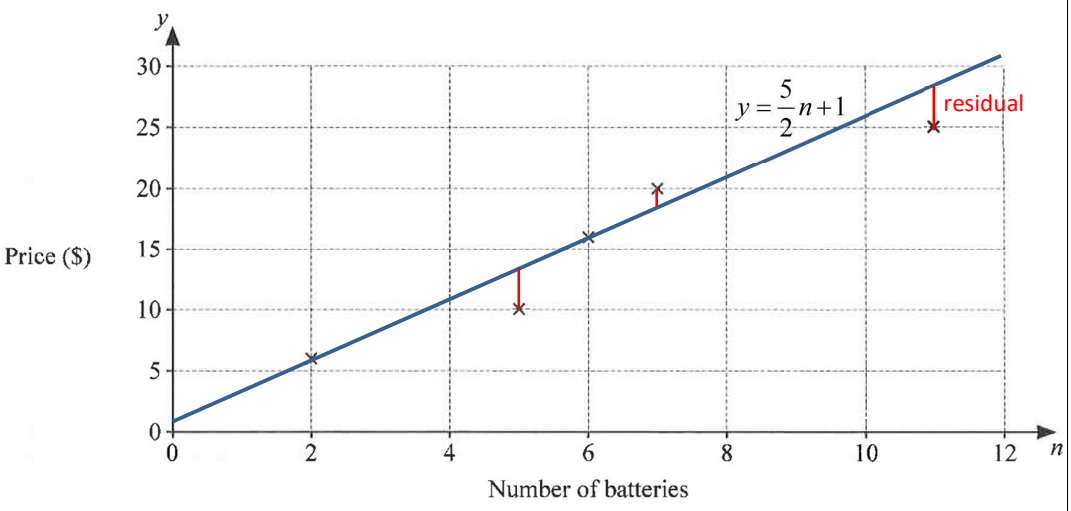
(cii) **[1]**	$Y \sim \text{B}(120, 0.03)$ Probability that, on a randomly chosen day, fewer than 3 burgers are found to have insufficient cheese $= \text{P}(Y < 3)$ $= \text{P}(Y \leq 2) \approx 0.29844 = 0.298$ (3 s.f.)	
(ciii) **[2]**	Let W be the number of days on which fewer than 3 burgers have insufficient cheese, out of 28 days. $W \sim \text{B}(28, 0.03)$ $\text{E}(W) = 28 \times 0.29844 = 8.36$ (3 s.f.)	
(iv) **[2]**	Let T be the number of burgers out of 3360 burgers that are found to have insufficient cheese in them. $T \sim \text{B}(3360, 0.03)$ Probability that, in a randomly chosen period of 28 days, more than 100 burgers are found to have insufficient cheese in them $= \text{P}(T > 100) = 1 - \text{P}(T \leq 100) = 0.506$ (3 s.f.)	

Question 10

No.	Suggested Solution	Remarks for Student
	Let X g be the mass of a plate before the holes are drilled. $X \sim \text{N}(200, 1.6^2)$.	
(a) **[1]**	Probability that the mass of a randomly chosen plate before drilling is more than 197.5 grams $= \text{P}(X > 197.5) \approx 0.94091 = 0.941$ (3 s.f.)	
(b) **[4]**	Let Y g be the mass of a drilled plate. Since $Y = 0.95X$, $\text{E}(Y) = 0.95\text{E}(X) = 0.95 \times 200 = 190$ $\text{Var}(Y) = 0.95^2 \text{Var}(X) = 0.95^2 \times 1.6^2 = 1.52^2$ $Y \sim \text{N}(190, 1.52^2)$. $\text{P}(190 < Y < 192) \approx 0.40588 = 0.406$ (3 s.f.) Let T be the number of drilled plates out of 8 drilled plates that have masses between 190 grams and 192 grams. $T \sim \text{B}(8, 0.40588)$ Probability that at least 5 of these 8 plates have masses between 190 grams and 192 grams $= \text{P}(T \geq 5) = 1 - \text{P}(T \leq 4) \approx 0.18291 = 0.183$ (3 s.f.)	

(c) **[3]**	The drilled plates are sold in packs of 20 randomly chosen plates. Let $K = Y_1 + Y_2 + Y_3 + Y_4 + ... + Y_{20}$ $\mathrm{E}(K) = 20\mathrm{E}(Y) = 20 \times 190 = 3800$ $\mathrm{Var}(K) = 20\mathrm{Var}(Y) = 20 \times 1.52^2 = 46.208$ $K \sim \mathrm{N}(3800, 46.208)$ Probability that the total mass of a pack of 20 drilled plates is less than 3805 grams $= \mathrm{P}(K < 3805) \approx 0.768997 = 0.769$ (3 s.f.)	
(d) **[4]**	Let S g be the mass of a screw. Then $S \sim \mathrm{N}(10, 0.3^2)$. Let B g be the mass of a bolt. Then $B \sim \mathrm{N}(44, 1.1^2)$. Let $V = K + S_1 + S_2 + ... + S_{80} + B_1 + B_2 + ... + B_{40}$ $\mathrm{E}(V) = \mathrm{E}(K) + 80\mathrm{E}(S) + 40\mathrm{E}(B) = 3800 + 80 \times 10 + 40 \times 44 = 6360$ $\mathrm{Var}(V) = \mathrm{Var}(K) + 80\mathrm{Var}(S) + 80\mathrm{Var}(B) = 46.208 + 80 \times 0.3^2 + 40 \times 1.1^2 = 101.808$ $V \sim \mathrm{N}(6360, 101.808)$ Let the mass exceeded by just 5% of the Value packs be m g $\mathrm{P}(V > m) = 0.05$ Therefore, m = 6376.596565 So, the mass exceeded by 5% of the Value packs is 6377 g (Note: Cambridge also accepted the answer 6376g)	

Question 11

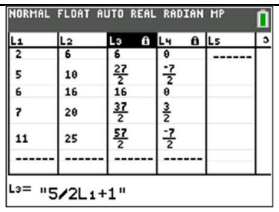
	Suggested Solution	Remarks for Student
(ai) [1]	As m increases, p increases at a decreasing rate.	
(aii) [4]	For $p = a + b\ln m$ product moment correlation coefficient, $r_1 = 0.928513224$ For $p = d + e\sqrt{m}$ product moment correlation coefficient, $r_2 = 0.9914381337$ Since r_2 is closer to 1 than r_1, $p = d + e\sqrt{m}$ gives the better fit with product moment correlation coefficient 0.991 (3 s.f.) As $p = 32.95256688 + 17.26980391\sqrt{m}$, $d = 33.0$ (3 s.f.) and $e = 17.3$ (3 s.f.) (3 s.f.)	
(aiii) [2]	If $m = 750$, $p = 32.953 + 17.270\sqrt{750} = 505.9114284$ The estimated price of the cereal is 510 cents. (nearest 10 cents) Since $r_2 = 0.991$ is close to 1 which suggests a strong positive linear correlation between m and p, and the value $m = 750$ is within the data range [24,5000] used to determine the linear model, the answer is reliable.	
(bi) [1] **(biiA) [1]**	 Note that the line passes through the points (2,6) and (6,16).	

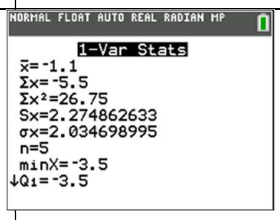
(biiB) [1]

Kai should use the sum of the squares of the residuals rather than the sum of the residuals when assessing the fit of the model because the residuals could be positive or negative and may cancel each other when they are added together while squares of the residuals are always positive and adding them would not result in any cancellation.

(biiC) [1]

Sum of the squares of the residuals $= 26.75$ or $\frac{107}{4}$.





(biii) [1]

The sum of the squares of the residuals for a line that is a better fit for the data will be ***less than*** the sum of the squares of the residuals for the line $y = \frac{5}{2}n + 1$ found in part **(bii)(C)**.

(biv) [3]

n	2	5	6	7	11
y	6	10	16	20	25
Residual $= y_i - \frac{5}{2}n_i - c$	$1-c$	$-\frac{5}{2}-c$	$1-c$	$\frac{5}{2}-c$	$-\frac{5}{2}-c$

For the line $y = \frac{5}{2}n + c$ to be a better fit, sum of the squares of the residuals must be less than $\frac{107}{4}$.

$$(1-c)^2 + \left(-\frac{5}{2}-c\right)^2 + (1-c)^2 + \left(\frac{5}{2}-c\right)^2 + \left(-\frac{5}{2}-c\right)^2 < \frac{107}{4}$$

$$2(1-c)^2 + 2\left(c+\frac{5}{2}\right)^2 + \left(c-\frac{5}{2}\right)^2 - \frac{107}{4} < 0$$

By considering the x-intercepts of the graph of $y = 2(1-c)^2 + 2\left(c+\frac{5}{2}\right)^2 + \left(c-\frac{5}{2}\right)^2 - \frac{107}{4}$,

$$-1.2 < c < 1$$

